

SOFTWARE REQUIREMENTS SPECIFICATION

Car Rental and Equipment Management System (CREMS)

Carpenters Motors and Carptrac MVP

Document purpose: This specification defines the agreed business, functional, data, security and quality requirements that will form the baseline for development, acceptance testing and customer sign-off.

Document control	Value
Client	Carpenters Fiji Pte Limited
Project	Car Rental and Equipment Management System (CREMS)
In-scope divisions	Carpenters Motors and Carptrac
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Classification	Client confidential - project use

Approval effect

Signing this document confirms the requirements baseline, not acceptance of a completed system. Changes after sign-off will be assessed through the project change-control process.

Document control

Version	Date	Prepared by	Description	Status
0.1	18 Aug 2026	CREMS team	Initial requirements consolidation	Draft
1.0	18 Aug 2026	CREMS team	Issued for Carpenters review and sign-off	For approval

Review and approval responsibilities

Role	Responsibility
Client sponsor	Confirms business scope, priorities, acceptance authority and change-control decisions.
Motors representative	Validates vehicle rental, customer, inspection, pickup/return and fleet data requirements.
Carp trac representative	Validates equipment categories, operators, meter rules, safety checks and maintenance requirements.
IT/security representative	Validates hosting, access, session, audit, backup, data-retention and security requirements.
CREMS project team	Maintains traceability, implements the approved baseline and records variances or assumptions.

Reference documents

- CREMS Feasibility Study, Version 1.0, 18 August 2026.
- CREMS Project Management Plan and twelve-week semester delivery baseline.
- Carpenters Vehicle Rental Inspection Checklist.
- Carp trac Equipment, Genset and Heavy Machine Rental Inspection Checklists.
- Rental Vehicle Stock List and Rental Stock List - CAT, supplied 18 August 2026.
- CREMS Client Data Mapping and Import Validation Pack, prepared for confirmation.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) records what CREMS must do, the business rules it must enforce, the information it must hold, and the quality and security standards it must meet. It is the controlling baseline for design, development, testing, user acceptance and handover during the CS400 Semester 2, 2026 project.

1.2 Product vision

CREMS will provide Carpenters Motors and Carptrac with one configurable web platform for customer access, asset registration and tracking, quotation and booking, approval, rental agreement, pre-hire and post-hire inspection, check-out/check-in, maintenance, invoicing and asset-level performance reporting. Division and branch boundaries must remain visible and enforceable throughout the system.

1.3 Definitions

Term	Definition
Asset	A vehicle, machine, generator, forklift or other item that Carpenters makes available for rental or hire.
Division	A Carpenters operating business configured in CREMS, initially Motors or Carptrac.
Branch	An operational location that may provide one or more division services.
Service	A configurable rental or hire offering belonging to a division.
Booking request	A customer's request for dates, service, category or asset; it is not confirmed until required checks and approvals pass.
Quotation	A versioned commercial offer containing rates, charges, tax, deposit, validity and approval status.
Hire / rental	A confirmed booking that has progressed through agreement and check-out into an active customer possession period.
Inspection	A configurable, signed pre-hire or post-hire condition record with readings, responses, notes and evidence.
Meter	Odometer kilometres, engine/operating hours, fuel level or another approved usage measurement.
MVP	The agreed Motors and Carptrac scope targeted for the semester delivery baseline.

1.4 Requirement language and priority

The word **must** denotes a mandatory requirement. **Should** denotes a high-value requirement that may be deferred only through approved change control. **Could** denotes a future or optional capability outside the committed MVP unless subsequently approved.

Priority	Meaning
Must	Required for the agreed MVP or for security, data integrity or legal/operational safety.

Priority	Meaning
Should	Important to business value; included where capacity permits and removed before any Must item.
Could	Future increment or enhancement; not part of the signed MVP baseline.

2. Stakeholders and user classes

User class	Primary responsibilities	Scope
Public visitor	Browse divisions, services, available categories/assets and indicative pricing; submit enquiries; register or sign in.	Public, customer-safe data only
Customer	Maintain account, request quotations/bookings, provide required information, accept documents, track status and view own records.	Own individual or corporate account
Super Administrator	Configure divisions, system-wide permissions, security settings and group-level access.	Group-wide
Administrator	Manage users, operational configuration and audit within delegated authority; cannot control a Super Administrator.	Assigned administrative scope
Branch Manager	Approve exceptions/high-value rentals and monitor branch operations and profitability.	Assigned division and branch(es)
Rental Officer	Manage customers, requests, quotations, allocations, pickup and return.	Assigned division and branch(es)
Maintenance Officer	Receive faults, schedule and complete work, record parts/labour/cost and release assets.	Assigned maintenance scope
Finance Officer	Manage invoices, payments, refunds, credit and financial reporting.	Assigned financial scope
Driver / Operator	View assignments, scan assets, record readings/timesheets, safety checks and delivery/pickup evidence.	Assigned tasks and assets

3. Scope baseline

3.1 In scope

- Carpenters Motors vehicle-rental operations and Carptrac equipment-hire operations.
- Public browsing plus authenticated individual and corporate customer accounts.
- Staff authentication, permission-based access and division/branch scoping.
- Configurable divisions, branches, services, asset categories, attributes, checklists, rates and approval workflows.
- Customer, asset, availability, quotation, booking, agreement, check-out, active hire and return workflows.
- Pre-hire and post-hire inspections based on the four supplied Carpenters checklists.
- Meter readings, QR identification, maintenance, costs, revenue, profitability, dashboards, reports, notifications and audit logs.
- An architecture that permits future divisions and hireable asset types to be configured without redesigning core modules.

3.2 Out of scope for this MVP

- Carpenters Shipping operations and assets, including bins, scaffolding and portable toilets.
- Freight forwarding, vessel operations, cargo manifests, customs, bills of lading and route optimization.
- Property leasing, hardware retail and other Carpenters divisions not explicitly approved for this baseline.
- Live bank/card payment-gateway integration, telematics/GPS hardware integration and accounting-system synchronization.
- Native mobile applications and guaranteed offline synchronization; the delivered interface is a responsive web application.
- Migration of unapproved production data. Supplied stock lists remain subject to mapping confirmation and cleansing.

Future-readiness rule: Out-of-scope divisions must be addable later through division, branch, service, category, attribute, pricing and checklist configuration. This requirement does not authorize delivery of those divisions in the current MVP.

3.3 Assumptions and dependencies

ID	Assumption / dependency
A-01	Carpenters will appoint representatives for Motors, Carptrac, IT/security and final acceptance.
A-02	Carpenters will confirm branch/location structure and the authoritative asset identifiers before import.
A-03	Customer, staff and operational data used for development will be synthetic or anonymised.
A-04	Email delivery depends on a valid approved SMTP/email-service configuration and sender domain.
A-05	Internet connectivity is available for normal operation; offline field synchronization is not committed.
A-06	Carpenters will provide timely decisions and UAT participants within the twelve-week schedule.
A-07	Rates, tax, deposits, credit limits and approval thresholds will be supplied or formally approved.

4. Functional requirements

Each requirement has a stable identifier for traceability. Acceptance criteria describe the minimum observable evidence required during testing or demonstration.

4.1 Identity, authentication and access

ID	Requirement	Acceptance criterion	Priority
IAM-01	Staff and customers must authenticate using a unique email address and password.	Valid credentials create a session; invalid credentials return a generic failure without revealing account existence.	Must
IAM-02	The system must securely hash passwords and must never store or log plain-text passwords.	Database and logs contain only framework-managed hashes; password values are absent from application logs.	Must
IAM-03	Accounts must support Invited, Activation Pending, Active, Locked, Suspended and Deactivated states.	Only Active accounts can sign in; every state change is audited.	Must

ID	Requirement	Acceptance criterion	Priority
IAM-04	Users must be able to request a time-limited, single-use password-reset link delivered to their registered email.	Expired, reused or invalid reset tokens are rejected; successful reset revokes prior sessions.	Must
IAM-05	Staff sessions must expire after a configurable inactivity period and absolute maximum lifetime.	Idle and absolute expiry tests end the session and require re-authentication.	Must
IAM-06	Session cookies must be HttpOnly, Secure and SameSite and must not expose authentication tokens to client scripts.	Browser inspection confirms cookie flags and no token in local/session storage.	Must
IAM-07	A login session must be bound to the originating browser-window session and invalidated when the window session is absent or replaced.	Copying the cookie into a different/new window context does not create a valid session under the agreed window-session design.	Must
IAM-08	The system must enforce permissions in addition to roles for sensitive actions.	API tests demonstrate allow/deny behaviour for configured permissions such as rentals.approve and reports.financial.	Must
IAM-09	Staff access must support one or more division/branch assignments, temporary access and group-wide access where authorized.	Records returned by APIs match active scope assignments and effective dates.	Must
IAM-10	Customer portal access must be separate from customer rental eligibility/business status.	Disabling portal access does not automatically block the customer from staff-managed rental eligibility, and vice versa.	Must
IAM-11	Authorized administrators must reset passwords, resend activation, lock/unlock accounts and revoke sessions.	Each action works within administrator authority and creates an audit event.	Must
IAM-12	Users must be able to view and revoke their active sessions.	Revoked sessions fail on their next authenticated request.	Should
IAM-13	The system must record successful/failed login, logout, expiry, reset and suspicious-login events with time, IP and device/browser summary.	Security history is queryable by authorized users and excludes secrets.	Must
IAM-14	Separation-of-duties rules must prevent self-elevation, unauthorized Super Administrator changes and self-approval of restricted transactions.	Negative authorization tests reject each prohibited combination.	Must
IAM-15	Sensitive Super Administrator and financial operations should support step-up verification/MFA when enabled.	Configured operations require a valid second factor; disabled configuration preserves the approved MVP login flow.	Should

4.2 Organization, divisions, branches and services

ID	Requirement	Acceptance criterion	Priority
ORG-01	Only Super Administrators must create, deactivate or materially configure divisions.	Normal administrators and branch managers receive 403 for division configuration APIs.	Must
ORG-02	A division must store code, name, description, contact details, branding, visibility, status, currency, tax and default rental terms.	Create/edit/view operations persist and return every approved field.	Must
ORG-03	A branch must store code, name, address, contacts, operating hours, pickup/return instructions, visibility and status.	Branch detail shows all configured operational fields.	Must

ID	Requirement	Acceptance criterion	Priority
ORG-04	A physical branch may serve multiple divisions and expose different services for each division.	Configuration enables Motors and/or Carptrac per branch without duplicate branch records.	Must
ORG-05	The division-branch relationship must control local contacts, manager, enabled services, booking availability, maintenance capability, pricing and workflow overrides.	A disabled relationship prevents new bookings while preserving history.	Must
ORG-06	Administrators must configure services without code changes, including hire unit, personnel/delivery requirements, quote requirement, documents, deposit, inspection and meter rules.	A new test service can be configured and used without application recompilation.	Must
ORG-07	Branches must support weekly hours, public holidays, temporary closures and pickup/return cut-off times.	Availability rejects or warns for a closed period according to configuration.	Should
ORG-08	Inactive divisions, branches and services must be hidden from new public/staff transactions but retained in historical records.	Historical reports still resolve names while selection lists omit inactive items.	Must
ORG-09	Branch managers may edit only approved operational details for their assigned branches and must not access the administration branch-configuration workspace.	UI navigation and API authorization both enforce the restriction.	Must
ORG-10	The platform must allow a future division and hireable asset category to be configured without redesigning core rental, asset and maintenance entities.	A configuration-only proof creates a sample inactive future division and category in staging.	Must

4.3 Customer and portal management

ID	Requirement	Acceptance criterion	Priority
CUS-01	The system must maintain separate individual and corporate customer accounts.	Registration and staff creation capture the correct fields and customer type.	Must
CUS-02	Customer records must hold a unique customer number, contacts, address, identification and active/blocked status.	Required-field and uniqueness validation is enforced.	Must
CUS-03	Corporate customers must support legal/tax details, multiple contacts, purchase orders, approved renters/drivers, credit limits and payment terms.	Corporate detail persists and displays all configured controls.	Should
CUS-04	Public visitors must browse customer-safe assets/services, indicative pricing and availability without staff-login links.	Public page shows no staff authentication entry and returns only approved public fields.	Must
CUS-05	A customer must authenticate before submitting a booking, tracking detailed status or viewing account documents.	Unauthenticated protected portal routes redirect to customer sign-in.	Must
CUS-06	Customers must select Motors or Carptrac and see a simplified experience relevant to the chosen service.	Vehicle customers are not forced through operator/equipment fields unless applicable.	Must
CUS-07	Customers must request quotations/bookings with dates, branch/service, category/asset preference and required contact information.	A valid request creates a reference and appears in the relevant staff queue.	Must
CUS-08	Customers must track their own quotations, bookings, agreements and invoices and must not access another customer's records.	Ownership authorization tests return 403/404 for cross-customer access.	Must
CUS-09	Customers must be able to submit enquiries and receive an acknowledgement/reference.	Valid enquiry persists and notification is queued/sent.	Should

ID	Requirement	Acceptance criterion	Priority
CUS-10	Staff must see a consolidated customer activity view across authorized divisions.	Customer profile shows scoped requests, rentals, invoices, cases and audit-relevant activity.	Should

4.4 Asset register, lifecycle and availability

ID	Requirement	Acceptance criterion	Priority
AST-01	Staff must register an asset with asset number, division, branch, service, category, manufacturer, model and active status.	Required fields are validated and the asset becomes searchable within scope.	Must
AST-02	Vehicle assets must support registration, VIN/chassis, engine, year, colour, odometer, registration/insurance and warranty information.	Vehicle profile displays configured values and uniqueness validation for approved identifiers.	Must
AST-03	Equipment assets must support serial, model, capacity/subtype, engine/operating hours and personnel requirement.	Equipment profile changes fields dynamically by configured category.	Must
AST-04	Super Administrators must configure asset categories and typed attributes including required, searchable, reportable and customer-visible flags.	A new attribute appears on the asset form and obeys its configured rules.	Must
AST-05	An asset profile must provide Overview, Specifications, Availability, Inspections, Meter Readings, Maintenance, Costs & Revenue, Documents, Lifecycle and Audit views.	Authorized staff can navigate each view from one asset workspace.	Must
AST-06	The system must generate a unique QR code that resolves to an authorized asset identification/check-in/check-out workflow.	Scanning a valid QR locates one asset; altered/unknown codes are rejected.	Must
AST-07	Asset lifecycle must support Commissioned, Available, Reserved, Pre-hire Inspection, Checked Out, On Hire, Returned, Post-hire Inspection, Maintenance/Damage Assessment and Retired/Disposed states.	Only valid transitions succeed and every transition records actor, time and context.	Must
AST-08	Lifecycle events must record user, timestamp, booking, previous/new status, meter, notes and evidence where applicable.	Asset history shows an immutable chronological record.	Must
AST-09	The system must store odometer, engine/operating hours, fuel and other configured meter readings with source and timestamp.	New readings cannot be lower than the prior approved reading unless an authorized correction is recorded.	Must
AST-10	Availability must consider bookings, maintenance, inspections, transfers, branch closures and personnel requirements.	Conflict tests prevent allocation for every blocking event type.	Must
AST-11	Staff and customers must receive a clear reason when an asset is unavailable and authorized staff should see suitable alternatives.	Availability response includes reason code and same-category alternatives where available.	Should
AST-12	Asset documents and photos must be associated with the asset and protected by authorization.	Unauthorized downloads fail; metadata records uploader, type and timestamp.	Should
AST-13	Asset financial data must include rental/service revenue and maintenance, operator, transport, labour, fuel and other costs.	Asset financial view reconciles stored charge/cost entries for a selected period.	Should
AST-14	The system should calculate utilization, gross profit, margin and cost per kilometre/hour where data is available.	Calculations use documented formulas and match test datasets.	Should
AST-15	Asset retirement/disposal must prevent future bookings while preserving history and financial records.	Retired assets disappear from availability but remain reportable.	Must

4.5 Quotation, booking and approval

ID	Requirement	Acceptance criterion	Priority
BKG-01	Staff must manage booking work queues for New Request, Quotation Required, Awaiting Approval, Confirmed and Cancelled/Expired.	Each queue shows count and only matching records within user scope.	Must
BKG-02	A booking workspace must show customer, service, dates, branch, asset/availability, pricing, approvals, documents, notes and activity.	Opening a request provides the complete workspace without requiring database access.	Must
BKG-03	The system must prevent overlapping confirmed/reserved bookings for the same asset.	Concurrent conflict test allows only one conflicting confirmation.	Must
BKG-04	A quotation must calculate hire duration and support base hire, operator/driver, transport, labour, fuel, deposit, discount and tax components.	Test quotation line items and totals match the approved calculation rules.	Must
BKG-05	Pricing must resolve the applicable rate by customer contract, branch override, division, category/asset and standard fallback order.	Pricing tests prove each precedence level and fallback.	Should
BKG-06	Rate configuration must support hourly, daily, weekly, monthly, per-unit, trip, kilometre, tonne, square-metre and fixed units.	Configured charge unit persists and calculates with entered quantity.	Must
BKG-07	Quotations must support version history, validity/expiry, approval status and customer acceptance/decline.	Revising a quote preserves prior immutable versions and creates a current version.	Must
BKG-08	Internal cost and margin information must be hidden from customers.	Customer quote/API contains selling amounts only.	Must
BKG-09	Approval workflows must allow configurable stage count, approver role/named user, value threshold, division scope and escalation.	A staged test booking routes to configured approvers in order.	Should
BKG-10	An approver must record approval, rejection or revision reason; unauthorized or self-approval must be rejected where prohibited.	Decision history records actor, time, stage and reason.	Must
BKG-11	Approval delegation should support effective start/end dates and division scope.	An active delegate can act only within the configured period/scope.	Should
BKG-12	A confirmed booking must require an available asset or approved category-allocation workflow, required approvals and valid customer status.	Confirmation fails with a clear unmet-requirement list when any condition is missing.	Must
BKG-13	The system must email approved quotations and booking confirmations using configured templates.	Delivery attempt and result are recorded against the transaction.	Must

4.6 Agreement, pickup, active hire and return

ID	Requirement	Acceptance criterion	Priority
HIR-01	A rental agreement must be generated only after booking confirmation, asset allocation, required identity/licence verification, payment/deposit checks and pre-hire inspection.	Agreement generation remains blocked until every configured prerequisite passes.	Must
HIR-02	Pickup must be a guided workflow: customer, QR/asset, identification/licence, meter/fuel, inspection, charges/agreement, signature and agent approval.	Staff cannot complete checkout while a mandatory step is incomplete.	Must

ID	Requirement	Acceptance criterion	Priority
HIR-03	The agreement must contain customer, authorized driver/operator, asset, dates, rates/charges, deposit, liability, condition acknowledgement and approved terms.	Generated document contains values from the confirmed booking and approved terms version.	Must
HIR-04	Customer and agent signatures must be captured at pickup and timestamped.	Signed agreement records names, signatures, time and completing user.	Must
HIR-05	After signing, the agreement must be immutable; corrections must use an addendum or controlled cancellation/reissue process.	Edit API rejects signed agreement changes and retains original document.	Must
HIR-06	Checkout must generate a PDF, email the customer copy, change the asset to On Hire and create audit/lifecycle records.	One completed checkout produces all four outcomes atomically or reports failure without partial state.	Must
HIR-07	Hire Operations must provide Pickup Today, On Hire, Due Today, Overdue, Return in Progress and Recently Completed queues.	Records appear in the correct queue based on status and timestamps.	Must
HIR-08	Return must be guided: QR scan, booking/customer confirmation, return time, meter/fuel, post-hire inspection, condition comparison, damage, final charges, acknowledgement and invoice.	Staff cannot finish return until required steps complete.	Must
HIR-09	The system must calculate approved late, fuel, cleaning, damage and excess-usage charges from return data.	Test returns match configured rules and retain calculation detail.	Must
HIR-10	The system must determine the next asset state from inspection/fault outcomes: Available, Inspection, Damage Assessment or Maintenance.	Return completion applies the documented decision rules without arbitrary staff status selection.	Must
HIR-11	Overdue hires must be clearly flagged and trigger configured alerts/notifications.	A past-due active hire appears as overdue and creates one deduplicated alert.	Must

4.7 Configurable inspections

ID	Requirement	Acceptance criterion	Priority
INS-01	Administrators must configure pre-hire and post-hire templates by asset category without code changes.	A new checklist item appears in the selected category/stage only.	Must
INS-02	Template items must support label, section, order, response type, required flag, evidence rule and issue action.	Configured properties are enforced on inspection completion.	Must
INS-03	Vehicle templates must support the supplied body, glass, lighting, tyres, interior, accessories, fuel and odometer fields.	All supplied vehicle checklist fields appear in the approved templates.	Must
INS-04	General equipment templates must support physical condition, controls, power, accessories, labels, operation, cables/hoses, cleanliness and damage.	All supplied general-equipment checklist fields appear.	Must
INS-05	Genset templates must support engine, control panel, battery, fuel, cooling, exhaust, alternator, canopy, meters, safety, electrical connections, earth stake, logbook, extinguisher, leaks and cleanliness.	All supplied genset checklist fields appear.	Must
INS-06	Heavy-machine templates must support engine, hydraulics, tracks/tyres, cab, alarms, attachments, leaks, extinguisher and damage diagram reference.	All supplied heavy-machine checklist fields appear.	Must

ID	Requirement	Acceptance criterion	Priority
INS-07	Inspections must capture customer/booking/asset context, meter/fuel, responses, notes, damage map, evidence, signatures, user and completion time.	Completed inspection record contains every applicable field.	Must
INS-08	Pre-hire and post-hire condition must be comparable item-by-item.	Return view highlights changed responses and new damage.	Must
INS-09	Safety-critical failures must block checkout or return-to-service according to approved configuration.	A configured critical failure prevents the relevant transition and explains why.	Must
INS-10	Approved issues must create a damage case or maintenance referral when configured.	Issue action creates one linked record and avoids duplicates.	Should
INS-11	Completed signed inspections must be immutable; authorized corrections must preserve an audit trail.	Direct edit is rejected; correction records original and replacement values/reason.	Must

4.8 Maintenance and operating cost

ID	Requirement	Acceptance criterion	Priority
MNT-01	Authorized staff must log preventive, corrective, inspection-referred and damage-related maintenance jobs.	Each job links asset, branch, type, fault, status, dates and reporter.	Must
MNT-02	Maintenance workflow must support Reported, Scheduled, In Progress, Waiting for Parts, Completed and Cancelled states.	Only documented transitions are accepted and audited.	Must
MNT-03	Preventive maintenance should be triggered by date, kilometres, engine/operating hours or whichever approved threshold occurs first.	Threshold test creates one due reminder/job at the boundary.	Should
MNT-04	Jobs must capture parts, internal labour hours/cost, external contractor invoices, transport, fuel and other actual expenses.	Completed job total reconciles its cost components.	Should
MNT-05	Maintenance must record downtime start/end and prevent conflicting rental allocation.	Unavailable period appears on asset calendar and blocks booking.	Must
MNT-06	Maintenance completion must record work performed, final cost, next-service rule and completion evidence.	Completion fails when required fields are missing.	Must
MNT-07	An asset must require authorized return-to-service approval before becoming Available after a blocking job.	Closing the job alone does not bypass approval.	Must
MNT-08	Maintenance history must show repeat failures and cumulative cost per asset.	Asset maintenance view aggregates jobs and identifies repeated fault categories.	Should
MNT-09	Authorized users should manage suppliers, parts inventory and parts usage.	Part issue reduces available quantity and links cost to the job.	Should
MNT-10	Warranty status and expiry should be visible when assessing maintenance cost responsibility.	Job view warns when the asset is within recorded warranty.	Should

4.9 Operators, drivers and field work

ID	Requirement	Acceptance criterion	Priority
OPS-01	Personnel records must support operator, driver, technician, labourer and supervisor types with branch/division scope.	Personnel selection is filtered by type and authorized scope.	Must

ID	Requirement	Acceptance criterion	Priority
OPS-02	Operator/driver qualifications must store certificate/licence, issue/expiry and safety-induction status.	Expired or missing mandatory qualification blocks assignment.	Must
OPS-03	Carptrac bookings must support required/optional operator assignment and availability checking.	Booking confirmation rejects a required service with no qualified available operator.	Must
OPS-04	Personnel assignments must record booking, asset, role, schedule, status and charge/cost rates.	Assignment appears in both booking and personnel schedule.	Should
OPS-05	Operators/drivers should record start/stop time, meter, safety checks, photos, signature and task status from a mobile-friendly workflow.	Authorized assigned user completes the workflow on a target mobile viewport.	Should
OPS-06	Timesheets must separate standard and overtime hours and support supervisor/customer approval where required.	Approved timesheet calculates charge and employee cost using configured rates.	Should

4.10 Finance, reporting and notifications

ID	Requirement	Acceptance criterion	Priority
FIN-01	The system must generate an invoice from confirmed rental charges and approved return adjustments.	Invoice lines reconcile quotation/booking and return charges.	Must
FIN-02	Authorized finance users must record payments, partial payments, refunds and outstanding balances.	Balance updates correctly and prohibited self-approval rules are enforced.	Must
FIN-03	Corporate accounts should support credit limits, payment terms, purchase orders and account statements.	Approval warns/blocks according to configured credit rule and statement reconciles invoices/payments.	Should
FIN-04	Asset profitability must compare rental/service revenue with maintenance, operator, transport, labour, fuel and other recorded costs.	Report matches a controlled test ledger per asset and period.	Should
FIN-05	Role-specific dashboards must show only useful, authorized operational and financial measures.	Rental officer, manager, maintenance, finance and administrator views differ and respect scope.	Must
FIN-06	Reports must include rental history, fleet/equipment utilization, maintenance, revenue, overdue rentals, customer history and asset profit/loss.	Each report filters by authorized date/division/branch and returns expected test data.	Must
FIN-07	Authorized reports should export to CSV, Excel and PDF.	Exported values match on-screen filters and totals.	Should
NOT-01	The system must send or queue booking confirmations, approved quotations, signed agreements, invoices, overdue notices and maintenance reminders.	Each configured event creates one notification with delivery status.	Must
NOT-02	Email templates must support approved placeholders and must not expose internal costs or unauthorized data.	Template preview and sent content contain only allow-listed fields.	Must
NOT-03	Failed delivery attempts must be logged and retryable without duplicating the business transaction.	A simulated failure records error and later retry succeeds once.	Must
RPT-01	The system must maintain an audit log for critical create, update, delete, approval, login, checkout, return, finance and configuration events.	Audit entry records actor, time, action, entity and non-secret summary.	Must

ID	Requirement	Acceptance criterion	Priority
RPT-02	Administrators must be able to search audit events but must not modify or delete them through normal application functions.	Audit mutation endpoints are absent or denied.	Must

5. Business rules

ID	Business rule
BR-01	Division and branch scope is enforced by the API; hiding a menu item is never sufficient authorization.
BR-02	An inactive division, branch, service, category, asset or account cannot be selected for a new transaction.
BR-03	One asset cannot have overlapping blocking bookings, maintenance, inspection or transfer periods.
BR-04	A meter reading normally cannot decrease; correction requires authority, reason and audit history.
BR-05	A required operator/driver must be qualified, available and assigned before the relevant booking is confirmed or checked out.
BR-06	A high-value or exception transaction follows its configured approval stages in order.
BR-07	The creator cannot approve their own restricted transaction when separation-of-duties configuration applies.
BR-08	A signed agreement and completed signed inspection are immutable records.
BR-09	An asset with a safety-critical failed inspection or open blocking maintenance job cannot be Available or checked out.
BR-10	Return completion derives the next asset state from inspection and maintenance rules.
BR-11	Customer-visible prices exclude internal cost, margin, staff comments and restricted asset/customer fields.
BR-12	Customer portal records are restricted to the authenticated customer's individual or linked corporate account.
BR-13	Tax, deposits, rate precedence, minimum hire, excess usage and after-hours rules are configuration-driven and versioned where used in a quote.
BR-14	Financial and profitability values are only as complete as recorded revenue and cost entries; incomplete data must be visibly identified.
BR-15	Production import requires approved mapping, corrected source data, staging reconciliation and written authorization.

6. Data requirements

The logical data structure must preserve group, division and branch scope while allowing flexible attributes and checklists. Exact physical schema may evolve provided these data obligations and relationships remain traceable.

Data area	Minimum information	Key controls
Identity/access	Users, roles, permissions, scope assignments, sessions, lifecycle state, login/security events	Least privilege; revocation; secrets excluded from logs
Organization	Divisions, branches, division-branch relationships, services, calendars, contacts, terms	Unique codes; effective active state; historical retention
Customers	Individual/corporate profile, contacts, identification, drivers, credit/business status, portal status	Customer ownership; sensitive-field restriction; duplicate detection
Assets	Identifiers, category, attributes, division/branch/service, lifecycle, location, ownership, value, documents	Approved identifier uniqueness; scope; status integrity
Meters	Asset, type, unit, reading, fuel, source, booking, timestamp, recorder	Monotonic validation; correction audit
Rental	Request, quotation versions/lines, approvals, booking, items, charges, agreement, pickup/return	Transactional consistency; immutable signed documents
Inspections	Template/version, stage, response, outcome, readings, evidence, damage map, signatures	Version retained; required/critical rules; immutable completion
Maintenance	Job, fault, status, dates, parts, labour, contractor, costs, downtime, evidence, return-to-service	Cost reconciliation; availability block; approval
Finance	Invoice, line items, payments, refunds, balances, customer/asset cost and revenue entries	Authorization; audit; calculation precision
Audit/notification	Event, actor, time, entity, summary; message template, recipient, result, retry	Append-only audit behaviour; no secret/sensitive content

6.1 Client-supplied data mapping status

The supplied stock lists contain 39 vehicle records and 79 Carptrac asset records. An anonymised 118-record mapping/import package has been prepared. Production identifiers have not been imported or committed to source control.

Import gate: Carpenters must confirm the canonical asset identifier, branch/location mapping, Carptrac Sort and Make codebooks, the meaning of Type, and identified vehicle data anomalies before production import.

6.2 Retention and privacy

- Collect only information required for rental, safety, maintenance, finance, access or audit purposes.
- Restrict identification, licence, contact, signature, financial and security-event information by role and purpose.
- Do not store API/SMTP secrets, passwords or raw authentication tokens in source control or logs.
- Carpenters must approve retention periods for agreements, inspections, photographs, invoices, audit records and login history.
- Deletion/anonymisation must not break statutory, financial, safety or audit retention obligations.

7. External interfaces

Interface	Requirement
Web user interface	Responsive React interface for current Chrome and Edge; customer portal mobile-friendly; staff optimized for desktop/tablet with mobile field flows where specified.
REST API	ASP.NET Core API using JSON, explicit authorization, validation, pagination/filtering and consistent error responses; no public administrative endpoint.
Database	SQL Server accessed through controlled application services/EF Core migrations; application users do not receive direct database access.
Email service	SMTP or approved email API for transactional notifications; configuration supplied through protected environment secrets.
QR scanner/camera	Browser-supported QR scanning or manual code entry; scanned code resolves through authorization rather than embedding sensitive data.
Document generation	PDF quotation/agreement/invoice outputs with stable references and authorized download/email delivery.
Future integrations	Accounting, payment, telematics and object-storage integrations must use versioned APIs/webhooks and remain outside MVP until approved.

8. Non-functional requirements

ID	Non-functional requirement	Verification
NFR-PERF-01	Ordinary authenticated API/page interactions must complete within 2 seconds for at least 95% of requests under the agreed pilot workload.	Performance test report
NFR-PERF-02	Standard reports should complete within 5 seconds for the agreed pilot dataset; long exports must show progress or run asynchronously.	Report timing tests
NFR-PERF-03	List APIs must use server-side pagination, bounded filtering and database indexes for common scope/status/date searches.	API and query review
NFR-SEC-01	Every non-public API endpoint must default to authenticated, explicitly authorized access.	Endpoint authorization inventory and negative tests
NFR-SEC-02	No cross-division, cross-branch or cross-customer data leakage is permitted.	Isolation test suite: zero leakage
NFR-SEC-03	Cookie-authenticated state-changing requests must be protected against CSRF and session fixation.	Security tests
NFR-SEC-04	Authentication and sensitive APIs must use rate limiting, lockout and generic error messages.	Abuse/lockout tests
NFR-SEC-05	All production/staging transport must use HTTPS with TLS 1.2 or later.	Deployment inspection
NFR-SEC-06	Secrets must be stored outside source control using approved secret/configuration mechanisms and rotated when exposed.	Repository and configuration audit
NFR-SEC-07	Uploaded documents/images must validate type and size and be protected from unauthorized access; production design should support malware scanning and private storage.	Upload security tests

ID	Non-functional requirement	Verification
NFR-SEC-08	Dependencies must be scanned before release with no unresolved Critical/High vulnerability accepted without documented approval.	Dependency scan report
NFR-SEC-09	The solution must address applicable OWASP Top 10 risks including injection, broken access, XSS, CSRF and insecure upload/configuration.	Security checklist and tests
NFR-REL-01	Critical multi-record operations such as checkout and return must be transactional and must not leave partial state.	Failure-injection integration tests
NFR-REL-02	Database backup and restore must be rehearsed before production acceptance.	Successful restore evidence
NFR-REL-03	Database migrations must have a tested apply path and documented recovery/rollback approach.	Migration rehearsal
NFR-REL-04	No unresolved Critical or High defect may remain at final acceptance unless the client signs a documented exception.	Defect register
NFR-USE-01	At least 80% of representative users must complete agreed core UAT tasks without facilitator assistance.	Observed UAT results
NFR-USE-02	Navigation must provide role-relevant menus, breadcrumbs, clear queue counts, one primary action and plain-language statuses.	Usability review/UAT
NFR-USE-03	Forms must provide field labels, validation messages, required markers and keyboard-accessible controls.	Accessibility/usability checklist
NFR-COMP-01	Critical journeys must pass on agreed current Chrome and Edge versions and responsive target viewports.	Cross-browser test results
NFR-MAIN-01	Code must avoid mutable global state, separate concerns, use reusable functions/services and follow repository conventions.	Peer review checklist
NFR-MAIN-02	All merges to the protected branch require peer review and relevant automated tests.	Repository protection/review evidence
NFR-MAIN-03	API, setup, configuration, operational and recovery documentation must be updated with implementation changes.	Definition-of-Done audit
NFR-SCALE-01	A new division, branch, service, category, attribute, rate unit and inspection template must be addable without core module redesign.	Configuration proof in staging
NFR-SCALE-02	The production sizing baseline must be confirmed with Carpenters for assets, customers, bookings, documents, concurrent users and retention volume.	Approved capacity worksheet
NFR-AUD-01	Critical audit events must be append-only through normal application functions and retained for the approved period.	Audit integrity and retention test

9. Reporting requirements

Report / dashboard	Minimum measures	Authorized audience
Rental Officer dashboard	Today's pickups/returns, inspections due, new requests, overdue hires and next action	Rental Officer; Branch Manager
Branch dashboard	Availability, utilization, overdue, approvals, maintenance, revenue/cost/profit for assigned branch	Branch Manager; authorized management

Report / dashboard	Minimum measures	Authorized audience
Division dashboard	Revenue, expenditure, utilization, downtime and asset profitability across division branches	Authorized division/group management
Asset profit and loss	Hire/service revenue, maintenance/operator/transport/labour/fuel cost, gross profit and margin	Finance; authorized managers
Rental history	Customer, asset, dates, branch, status, charges and agreement reference	Scoped staff; customer sees own history
Fleet/equipment utilization	Available, reserved, on hire, inspection, maintenance, downtime and utilization percentage	Operations and management
Maintenance	Jobs, due/overdue schedule, faults, downtime, parts/labour/contractor cost and repeat failures	Maintenance and management
Overdue rentals	Customer/contact, asset, due date/time, days overdue, balance and follow-up status	Rental/branch/finance staff
Customer rental history	Requests, quotes, rentals, incidents, invoices, payments and balances	Scoped staff; customer sees own account
Audit/security	Critical actions and security events filtered by actor/date/entity/action	Administrator/Super Administrator

10. Acceptance and traceability

10.1 Requirement acceptance

A requirement is accepted only when its implementation is demonstrated in the agreed environment, its acceptance criterion passes, applicable authorization/isolation tests pass, and the result is recorded in the UAT or test evidence. A screen mock-up or partial backend endpoint alone does not constitute acceptance of an end-to-end requirement.

10.2 Minimum end-to-end UAT scenarios

Scenario	Expected journey
UAT-01	Individual customer browses Motors, signs in, requests a vehicle, receives quotation/confirmation and tracks status.
UAT-02	Corporate Carptrac customer requests equipment requiring an operator, purchase order and approval.
UAT-03	Rental officer allocates an available asset and completes pre-hire inspection, agreement signature and QR checkout.
UAT-04	Branch manager approves an authorized high-value/exception transaction and cannot approve a prohibited own transaction.
UAT-05	Operator views assignment, records safety/meter/timesheet data and remains limited to assigned scope.
UAT-06	Rental officer processes return, compares inspections, records damage/charges and routes the asset to maintenance.
UAT-07	Maintenance officer records work, costs and return-to-service; availability is blocked until approved.

Scenario	Expected journey
UAT-08	Finance officer generates invoice/payment and branch manager reviews scoped asset profitability.
UAT-09	Unauthorized cross-branch, cross-division and cross-customer API requests are rejected.
UAT-10	Session timeout, reset, revocation and security-event history behave as approved.

10.3 Requirements traceability

The project team will maintain a traceability register linking each approved requirement ID to design/component, implementation task, test case, test result, defect (if any) and client acceptance status. Deferred or changed requirements must retain their original ID and a recorded disposition.

11. Open decisions requiring Carpenters confirmation

ID	Area	Decision required	Owner	Required by
DEC-01	Assets	Which identifier is the official permanent asset number for Motors and Carptrac?	Fleet/asset owners	Before production import
DEC-02	Branches	Are Suva Rona Street, Foster Road and Autofield Nakasi separate branches, depots or sub-locations?	Operations	Before organization setup approval
DEC-03	Carptrac taxonomy	Provide definitions for all Sort and Make codes and confirm whether Type means model, capacity or subtype.	Carptrac asset owner	Before category/data approval
DEC-04	Vehicle data	Confirm Stocked_Date meaning and correct missing/invalid registration, VIN and suspicious odometer values.	Motors fleet owner	Before import
DEC-05	Inspections	Which checklist items are safety-critical, require photos, block checkout/availability or create maintenance/damage cases?	Rental, HSE, maintenance	Before template acceptance
DEC-06	Signatures	Confirm required signatories for each vehicle/equipment delivery and return scenario.	Rental operations	Before agreement/inspection acceptance
DEC-07	Fuel/meters	Confirm fuel scale and approved primary/secondary meter types by category.	Operations/maintenance	Before meter acceptance
DEC-08	Pricing	Provide authoritative rates, tax treatment, deposits, discounts, minimum periods, excess usage and approval thresholds.	Finance/management	Before quotation acceptance
DEC-09	Security	Confirm session idle/absolute timeouts, MFA policy and authorized customer/staff retention periods.	IT/security	Before security acceptance
DEC-10	Capacity	Confirm expected asset/customer/booking/document volumes and concurrent users for production sizing.	IT/management	Before deployment design
DEC-11	Email	Confirm production sender domain, service, recipients and retry/escalation policy.	IT/communications	Before email acceptance

ID	Area	Decision required	Owner	Required by
DEC-12	Hosting	Confirm staging/production hosting ownership, backup location, recovery objectives and support handover model.	IT/management	Before deployment

Sign-off treatment: Open decisions do not authorize the CREMS team to invent business rules. They remain controlled clarifications. Where implementation must begin earlier, the team will use an explicitly documented provisional assumption and return it for approval.

12. Change control

1. Requestor records the proposed change, business reason, urgency and affected requirement IDs.
2. CREMS team assesses scope, design, data, security, testing, schedule and resource impact.
3. Client sponsor and project/academic authority approve, reject or defer the change.
4. Approved change receives a new document version, traceability update and revised acceptance criteria.
5. Development begins only after approval unless the change is an urgent security correction within existing authority.

Verbal discussion does not change the signed baseline until the decision is recorded and approved.

13. Client sign-off

By signing below, the parties confirm that this document accurately represents the requirements baseline for the CREMS Motors and Carptrac MVP. Sign-off authorizes the project team to proceed against this baseline and subjects later additions, removals or material changes to change control. Sign-off does not state that development is complete or that final acceptance has occurred.

Approval statement	Status / comment
In-scope divisions and exclusions are correct.	
User classes, access boundaries and security requirements are acceptable.	
Functional requirements and priorities represent the agreed MVP.	
Supplied stock-list mappings and inspection templates may proceed to detailed validation, subject to open decisions.	
Non-functional requirements and UAT approach are acceptable.	
Open decisions and responsible owners are acknowledged.	

13.1 Signatures

Role	Name	Signature	Date
Client Sponsor - Carpenters Fiji	Amit Kumar	_____	_____
Motors Representative		_____	_____
Carptrac Representative		_____	_____
Client IT/Security Representative		_____	_____
Academic Supervisor	Dr Ravneil Nand	_____	_____
Project Leader	Kavish Chandra	_____	_____

Appendix A - Requirements summary

Area	Total	Must	Should
Identity, authentication and access	15	13	2
Organization, divisions, branches and services	10	9	1
Customer and portal management	10	7	3
Asset register, lifecycle and availability	15	11	4
Quotation, booking and approval	13	10	3
Agreement, pickup, active hire and return	11	11	0
Configurable inspections	11	10	1
Maintenance and operating cost	10	5	5
Operators, drivers and field work	6	3	3
Finance, reporting and notifications	12	9	3
Non-functional requirements	26	26	0

Detailed totals are informational; the signed priority in each requirement row controls.